

Calculus (PR12)

Week 4: Indefinite Integration Techniques

Learning Goals



Antiderivatives

- Understand the **relationship** between **differentiation** and **antiderivatives**
- **Compute** basic antiderivatives



Substitution

- Use **substitution** to evaluate integrals
- Understand the rule as the reverse of **chain rule**



Integration by parts

- Apply **integration by parts** to integrate **product functions**



Think and Share

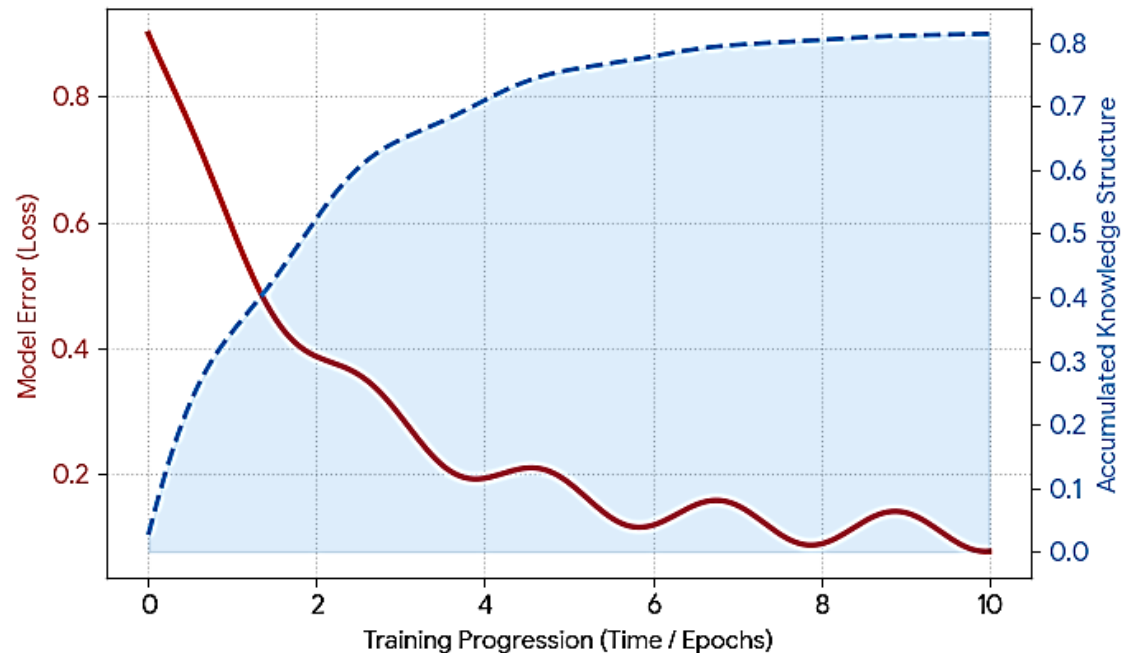
If differentiation measures instantaneous change...

- **What reconstructs the whole story?**
- **What is the mathematics of accumulation?**

Today's big idea: How do intelligent systems combine many tiny signals?

Why This Matters in AI

- **Neural networks**
- **Probability distributions**
- **Expected values**
- **Transformers**



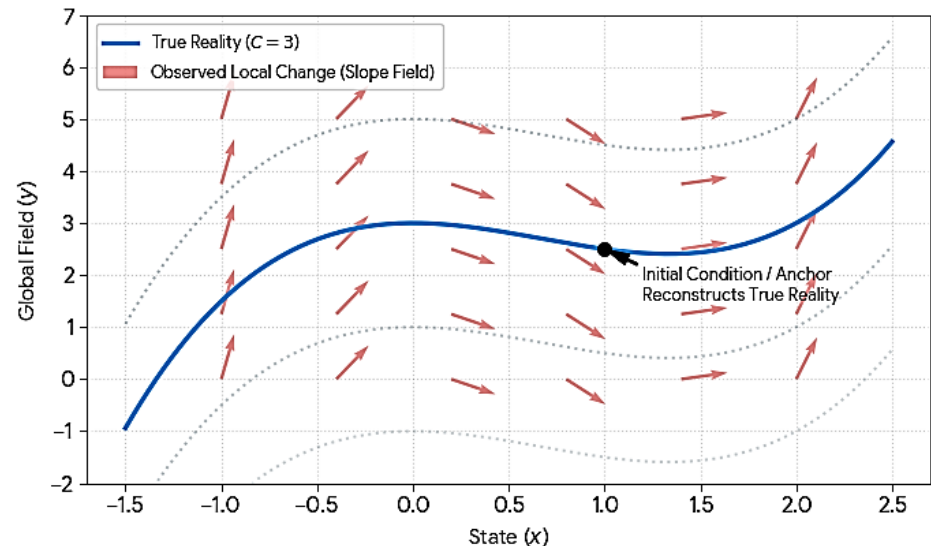
Modern AI is deeply built on accumulation.



The Deep Idea of Antiderivatives

- Antiderivatives reconstruct hidden structure
- Integration reverses local observation into global understanding
- Differentiation removes some information

Question: Could you reconstruct reality if all you knew was change?

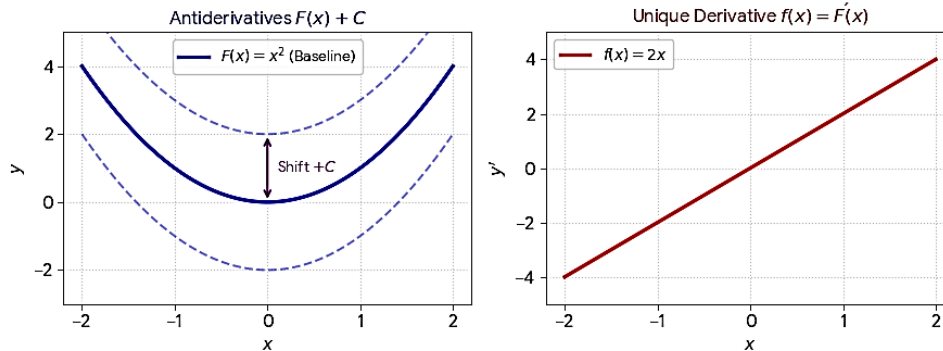


Core Mathematical Structure

- **If $F'(x) = f(x)$, then:**

$$\int f(x) dx = F(x) + C$$

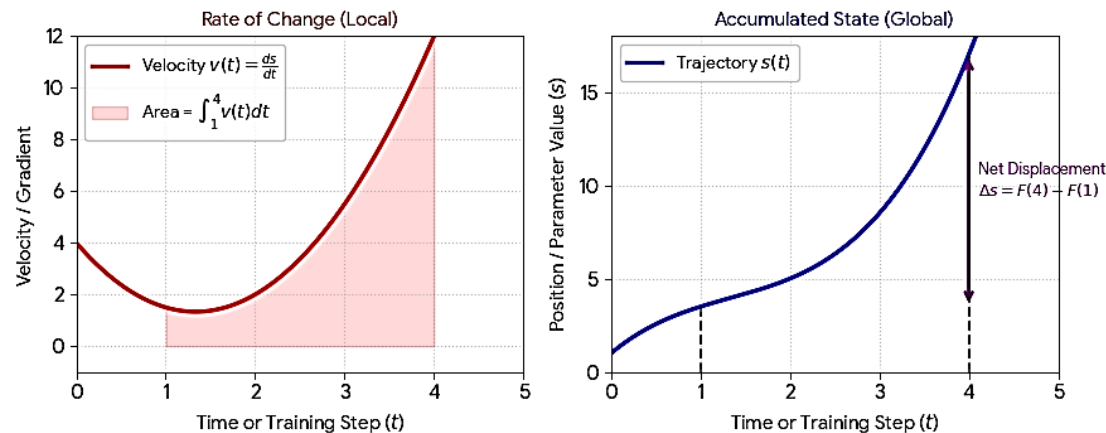
- **The constant matters because differentiation erases constants.**



Question: What information is lost when we differentiate?

Area Accumulation and Gradient Reversal

Discussion point: If velocity is known, what does integration recover?



In AI, accumulated gradient updates shape learning trajectories.



Pattern Instead of Memorization

To be successful:

- Recognize families of functions
- Polynomials, exponentials, logarithms, trigonometric forms
- Experts do not memorize randomly; they recognize structure

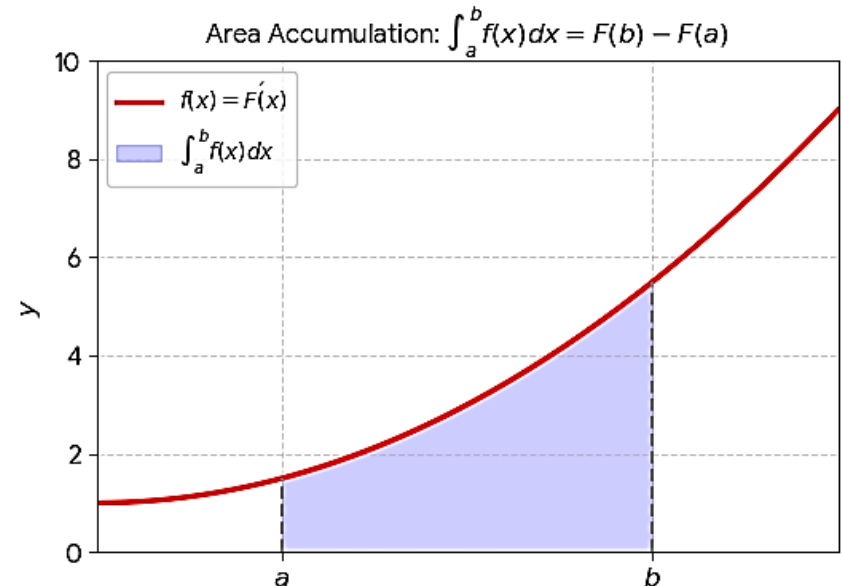
Important question to ask: What pattern do I see first?



The Fundamental Theorem of Calculus

If $F'(x) = f(x)$, then:

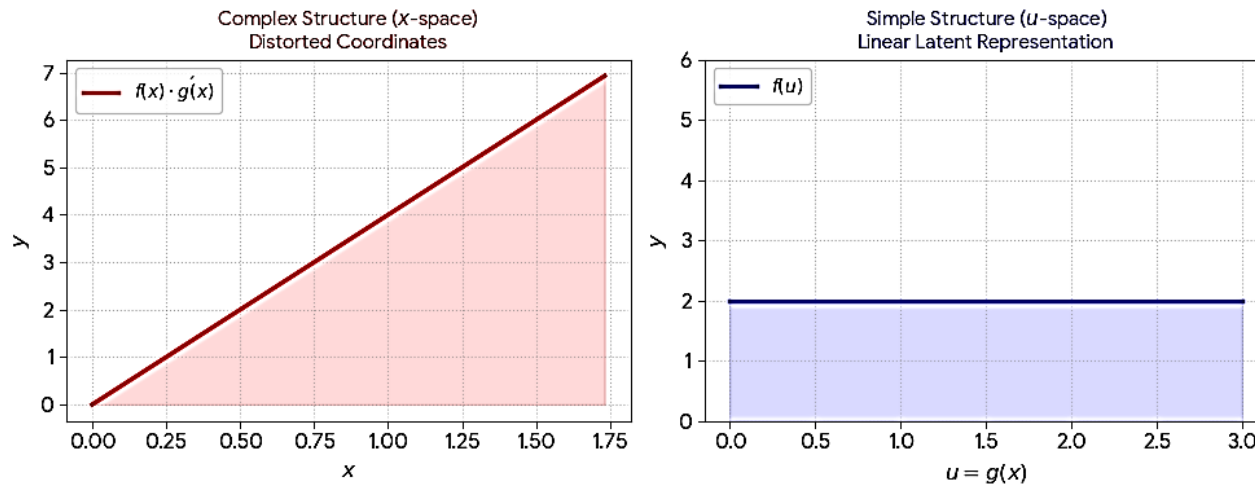
$$\int_a^b f(x) dx = F(b) - F(a)$$



Question: This theorem is computationally revolutionary, why?

Substitution as Representation for Learning

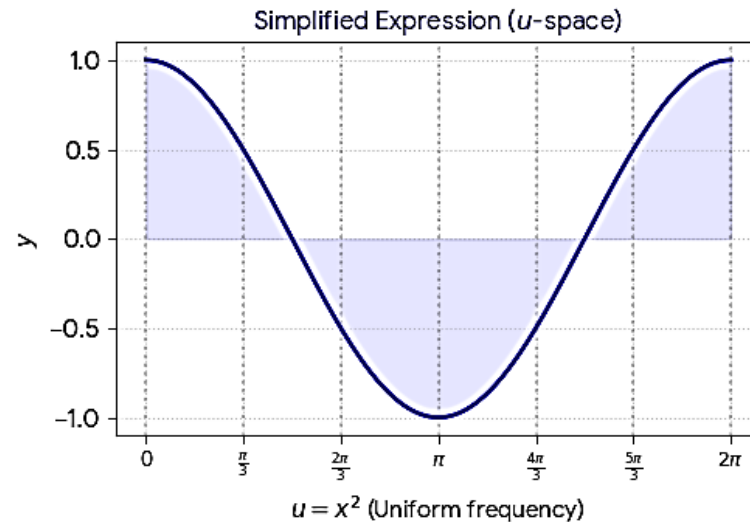
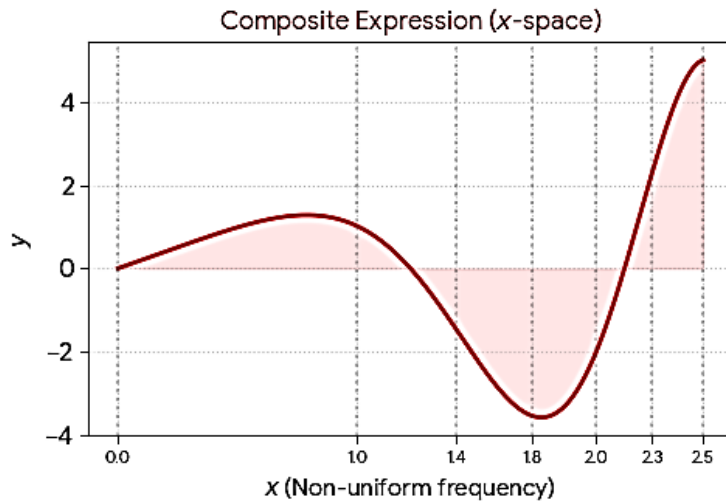
- **Substitution changes perspective**
- **We simplify problems by moving to better coordinates**
- **This mirrors latent representations in machine learning**



In AI, changing representations often creates simplicity.



Substitution: The Chain Rule in Reverse



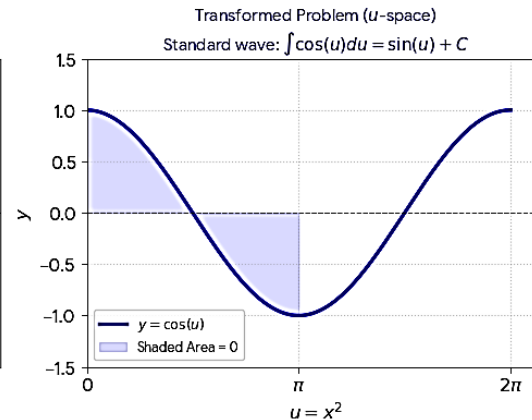
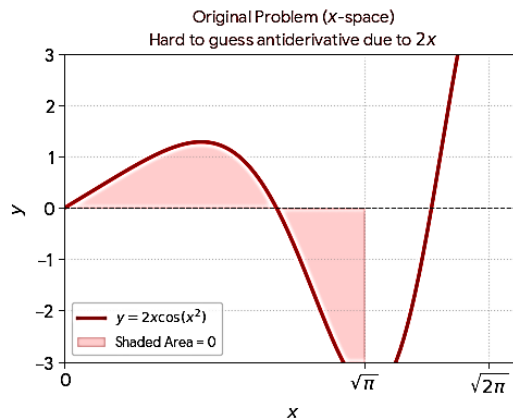
Goal: Simplifying composite expressions.

Substitution Example

$$\int 2x \cos(x^2) dx = ?$$

Let $u = x^2$ then $du = 2x dx$ and integral becomes:

$$\int \cos(u) du = \sin(u) + C = \sin(x^2) + C$$

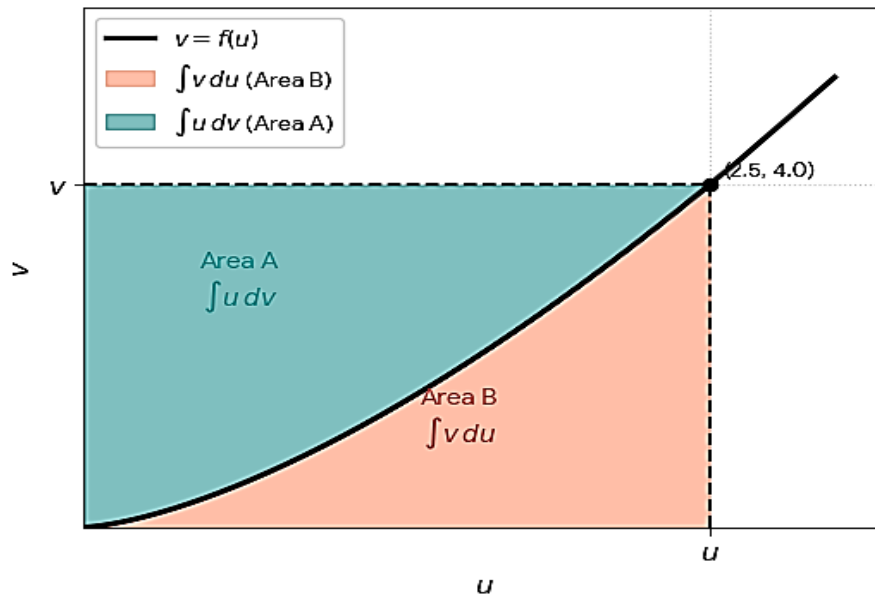


Question: How did we detect the hidden structure?



Integration by Parts

- Some problems become easier when complexity redistributed
- Integration by parts reverses the product rule
- It trades one difficulty for another



$$\int u dv = uv - \int v du$$

Integration by Parts - Example

Compute $\int x e^x dx$.

Assume $u = x$ and $dv = e^x dx$, then $du = dx$ and $v = e^x$.
The result will then be: $x e^x - e^x + C$

Question: Why was this choice strategic?

$$\int u dv = uv - \int v du$$

Set $u = x$
(Algebraic $\rightarrow$ Easy to differentiate)

Set $dv = e^x dx$
(Exponential $\rightarrow$ Easy to integrate)



$du = 1 \cdot dx$
(Completely kills the variable x !)

$v = e^x$
(Remains clean and stable)

Choosing u and dv

Heuristic **LIATE**:

- **L**ogarithm
- **I**nverse trigonometric
- **A**lgebraic
- **T**rigonometric
- **E**xponential

Question: Why choose logarithm first?



Challenge

Which method would you use?

- $\int x \cos(x) dx$
- $\int x e^{x^2} dx$
- $\int \ln(x) dx$

Do not compute first - identify the structure first.



Common Mistakes

- **Forgetting $+C$**
- **Incorrect substitution**
- **Choosing difficult u in parts**
- **Confusing definite and indefinite integrals**

Conceptual Wrap-UP

Note:

- Antiderivatives reverse change
- Substitution simplifies structure
- Integration by parts redistributes complexity

AI take away:

Modern Machine Learning continuously manipulates accumulated quantities.

Next week:

Definite integrals and differential equations.

?

Questions

- **Follow-up reading: Calculus: A Complete Course, (sections 2.10 & 5.6 & 6.1)**
- **Joining the Discussion Groups and the Lab**



Substitution in Modern AI

- Normalizing flows transform probability spaces
- Variational inference changes coordinate systems
- Embeddings compress complex structure into simpler geometry
- Expected values often require integration by parts

